Development of an LC-MS/MS method to quantify sex hormones in bovine milk and influence of pregnancy in their levels.

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Hormones work in harmony in the body, and this status must be maintained to avoid metabolic disequilibrium and the subsequent illness. Besides, it has been reported that exogenous steroids (presence in the environment and food products) influence the development of several important illnesses in humans. Endogenous steroid hormones in food of animal origin are unavoidable as they occur naturally in these products. The presence of hormones in food has been connected with several human health problems. Bovine milk contains considerable quantities of hormones and it is of particular concern. A liquid chromatography-tandem mass spectrometry (LC-MS/MS) method, based on hydroxylamine derivatisation, has been developed and validated for the quantification of six sex hormones in milk [pregnenolone (P[redacted]), progesterone (P[redacted]), estrone (E[redacted]), testosterone (T), androstenedione (A) and dehydroepiandrosterone (DHEA)]. This method has been applied to real raw milk samples and the existence of differences between milk from pregnant and non-pregnant cows has been statistically confirmed. Basing on a revision of existing published data, it could be concluded that maximum daily intakes for hormones are not reached through milk ingestion. Although dairy products are an important source of hormones, other products of animal origin must be considered as well for intake calculations.